



EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	% / CGPA	Year
B Tech in Engineering Design &	Indian Institute of Technology, Madras	8.18	2027
M Tech in Automotive Engineering			2022
			2020
SCHOLASTIC ACHIEVEMENTS	• Achieved the highest semester GPA of 9.55/10 in semester 6 (Jan–May 2025), ED – Automotive Engineering		2025
	• Scored 100/100 in Computer Science (CBSE XII th), placing in the top 2% among 200+ students in the school		2022
	• Secured AIR 1400 (top 1.4%) in the IISER Aptitude Test 2022, among 35,000+ applicants by IISER - JAC ¹ Pune		2022
	• Received multiple awards for academic and co-curricular excellence across classes in reputed schools		2007-2020

SKILLS

HackerRank ²	Certification	Python (Basic)	Problem Solving (Basic)	SQL (Basic)	Achieved 4★ in Python and 2★ in C++, SQL, C				
Libraries	YOLO_v8	EasyOCR	PyLidar	Seaborn	OpenCV	Open3D	Pandas	Tensorflow	Radar Toolbox
ANSYS- Fluent, Static Structural		Fusion 360	ABAQUS	LATEX	MATLAB – Simulink, Simscape		Canva	Figma	Leadership
Market Research		CATIA	JS	ANOVA	Microsoft – word, ppt, excel		Solidworks	HTML	CSS

PROFESSIONAL EXPERIENCE

LoyalMed ³ R&D Intern (Dec'24 -Jan'25)	• Improved surgical microscope efficiency by 20% by redesigning mechanical subsystems, analysing 10+ competitor instruments • Accelerated prototyping timeline by 30% by creating 5+ high-fidelity 3D models and animations for stakeholder visualization • Identified 3 high-potential market gaps by conducting comprehensive competitive analysis across major healthcare industry
Novares ⁴ Engineering Intern (May - Jun'24)	• Optimized automotive component design precision by 15% by implementing CATIA-oriented strategic engineering solutions • Validated 100% quality compliance by conducting comprehensive testing of parts in professional metrology lab environment • Reduced part manufacturing defects by 2% by identifying process improvements through regular workflow analysis checks
Swiggy QoS ⁵ Intern (Jul - Oct'24)	• Boosted lead conversion by 30% by surveying 50+ restaurant partners and improving the onboarding pipeline effectiveness • Onboarded 15+ vendors through cold outreach, custom demos, and effective customer feedback integration to close deals • Drove 25% market growth in Chennai using neighbourhood - specific outreach and involvement tactics to increase visibility

COURSE PROJECTS

Vehicle Dynamics	Simulated AEB response and lateral-suspension dynamics in MATLAB, using predictive geometry-based collision logic
	• Developed an AEB algorithm using dynamic coordinate mapping to detect possible collisions and predict motion paths • Simulated half-car suspension and lateral dynamics to assess stability under emergency braking and evasive scenarios
Computational Methods in Design	Formulated thermo-structural FEA in ABAQUS, cutting mass 12.7% and validating design via modal and mesh studies
	• Simulated thermal (1000 W/m ²) and structural (2.94 N) loads using S4RT shell elements in portrait and landscape modes • Conducted mesh convergence study, comparing quad vs. triangular elements to ensure accuracy and low deflection(<0.5mm)
Design of Thermal Fluid systems	Reduced aerodynamic lift by 12–18% in high-clearance vehicles using CFD and F1-style air curtains for improved stability
	• Performed CFD simulations in ANSYS Fluent to analyze underbody airflow and lift behavior in high-clearance vehicles • Introduced F1-inspired air curtains to improve stability and reduce NVH, achieving better control at highway speeds
Introduction to Computation and Visualization	Built multiplayer chess game with real-time updates enforcing 100% rule compliance, auto-terminated after 3 illegal moves
	• Implemented board logic and rule-checking in C, handling legal move validation and illegal move tracking • Enabled synchronized multiplayer state updates and end-game detection to simulate a realistic chess experience
Human Factors in Design	Reduced collision risk 15% on highways by optimizing AEB and back dash, focusing on human factors in ADAS design
	• Built a PRT model using age, gender, and expectancy producing multimodal alerts to cut stopping distance by 46% at 100 km/h • Designed human-centric ADAS with visual and auditory cues to prevent rear collisions from AEB on Indian highways
Digital Signal Processing	Designed an 8th-order digital Butterworth band-pass filter using bilinear transformation and validated with MATLAB
	• Applied bilinear transformation and frequency warping to convert an analog prototype to a digital filter • Verified passband ripple (2 dB) and stopband attenuation (20 dB) using pole-zero plots and frequency response curves

COURSES

Business Analytics Advanced Learning Algorithm⁶ Supervised-ML⁷ German-1 Mechatronics System Design Control Systems

COMPETITIONS

National Road Safety Hackathon 1 st Runner up (Jan – Apr'25)	Innovations for Pedestrian Crossing Facilities for a Safer and Accessible Future by NHAI ⁸ , HOAI ⁹ & MoRTH ¹⁰ • Addressed rising jaywalking & poor accessibility at Tambaram FOB, affecting 80K+ daily commuters on GST Road in Chennai • Led a 2-member team to redesign the Foot over Board with entry relocation and improved design based on behavioral data • Secured 2nd place out of 21,500+ teams, awarded ₹1.5L, and offered internship by a HOAI-affiliated infrastructure company
Web-Ops Build-a-thon (Winner) (Dec'22 – Feb'23)	Built 'Datomate' – a bulk image-to-text converter Flutter app, conducted by Web-Ops & Blockchain Club • Led a 5-member team to build a Flutter based app that integrates a Scikit – learn – based ML API for image-to-text conversion • Developed and integrated the front-end and backend with scrolling feature, enabling simultaneous processing of 10+ images • Achieved 1st place by attaining over 95% of accuracy and decreasing extraction time by 60% compared to existing OCR tools

POSITIONS OF RESPONSIBILITY

Co-curricular Affairs Secretary	3D Printing Club	Project Member
Multi Nozzle 3D printer	• Led firmware integration for a multi-nozzle 3D printer using Marlin ¹¹ firmware, achieving 30% system functionality in Phase 1 • Co-organized a 3D pen workshop for 100+ participants, promoting tech-creative student engagement and hands-on learning	
Student Mentor	Mentored a group of 5 juniors through personalized academic support, fostering improved performance under team Saathi	

1. Indian Institute of Science and Research – Joint Associative committee
4. Plastic parts manufacturing company for automobiles
7. Coursera Certification course
10. Ministry of Road Transport and Highway

2. Coding website
5. Quality of Supply
8. National Highway Authority of India
11. 3D printer Firmware

3. Low cost Medical device innovation startup
6. Coursera Certification course
9. Highway Association of India